Matheus Ribeiro Vidal

Mechanical Simulation Engineer / Aerodynamicist

Munich, Bavaria, Germany

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CFD engineer and aerodynamicist with 3+ years in multiphysics simulation (Prysmian: structural-thermal-electromagnetic coupling for subsea systems), aerodynamic research (UnB: ground-effect wing design), and ongoing TUM Computational Mechanics MSc. Specialized in turbulence modeling (RANS/LES), CFD workflow automation, and validation. Proficient in ANSYS, OrcaFlex, SU2, MGLET, and Python-based simulation ecosystems.

TECHNICAL SKILLS

CFD Methods

RANS (k-ω SST, k-ϵ), LES (SIGMA), structured and unstructured mesh generation, y+ analysis, finite volume/element methods

Solvers & Tools

ANSYS, FEA, OrcaFlex, Altair Flux, MGLET, SU2, OpenFOAM, Gmsh, SALOME, Paraview

Programming

Python (NumPy, SciPy, Matplotlib), MATLAB, C++, Bash, Git/GitHub, Linux, HPC

Languages

Portuguese (fluent), English (fluent), German (B2), Spanish (intermediate)

INTERESTS

- Motorsports
- Automotive
- Aerospace
- Offshore & Marine
- Renewable Energy
  - Wind & Solar
  - Nuclear
- CFD Research

EDUCATION

Technische Universität München / MSc. Computational Mechanics
2024–Present · Munich, Germany

- Coursework: Fluid Dynamics & Turbulence Modeling, Structural Mechanics, Finite Element Modelling & Simulation, Computational Material Modeling, Isogeometric Analysis and Design.
- Projects: SIGMA LES validation (MGLET, 15M+ cells vs DNS); brake-noise mode-coupling analysis (MATLAB); non-linear material implementation (IGAeasy).

University of Brasília / BSc. Mechanical Engineering
2017–2023 · Brasília, Brazil

- Thesis: Full aerodynamic development pipeline: geometry generation, CFD setup, turbulence modelling, and downforce performance analysis for a multi-element wing in ground effect.

WORK EXPERIENCE

Technische Universität München / Research Assistant / LES Turbulence Modeling (HiWi)
Sep 2025 to May 2026 · Associate Professorship of Hydromechanics – Munich, Germany

- Validate SIGMA LES turbulence model in MGLET against DNS benchmark to assess subgrid closure solver accuracy.

Prysmian / Mechanical Simulation Engineer / CAE R&D
Sep 2023 to Sep 2024 · Methodology and Simulations R&D Dept. – Vitória, Brazil

- Engineered structural and dynamic simulations for submarine umbilical cable systems (up to 3 km length, depths 300 - 1500 m) per ISO 13628-5 and Petrobras I-ET-3000.00-1519-29B-PZ9-004; coupled OrcaFlex, ANSYS, and Altair Flux for multiphysics validation.
- Developed Python scripts and tools to automate simulation workflows and client reporting; contributed to several in-house automation utilities.

University of Brasília / Undergraduate Researcher
Aug 2022 to Aug 2023 · CAALab (Computational Aeroacoustics Lab) – Brasília, Brazil

- 2nd place, Best Undergraduate Research (FAP-DF, 2023): validated open-source CFD workflow via 20 RANS simulations at Re approx 856k against wind-tunnel experiments on a multi-element wing in ground effect.

PUBLICATIONS

- **Numerical Exploration on the Effect of the Hydrodynamic Drag Coefficient Upon Topside Loading in Steel Tube Umbilical Risers** — *IV Congresso Brasileiro de Fluidodinâmica Computacional (CBCFD 2024)*, September 2024, Vitória, Brazil
  M. R. Vidal & F. C. Coutinho
  DOI: 10.17648/cbcfd-2024-192052
- **Numerical Simulation of Turbulent Flows over Multi-Element Wings in Ground Effect for Downforce Generation** — *XII Congresso Nacional de Engenharia Mecânica (CONEM 2024)*, July 2024, Natal, Brazil
  M. R. Vidal, A. M. Giolo & B. G. Pimenta (Universidade de Brasília)
  DOI: 10.26678/abcm.conem2024.con24-0426

RECOMMENDATIONS

Marianna Broseguini
marianna.broseguini@prysmian.com Product Development Coordinator, Prysmian · Led departments of Simulations & Methodology, Product Qualification, Prototype Manufacturing, and Experimental R&D

Filipe Cestari Coutinho, M.Sc.
filipecestaric.professional@gmail.com Simulations & Methodology Team Leader, Prysmian · Senior engineer collaborating on structural, dynamic, thermal, electromagnetic simulations and workflow automation

Prysmian Group

PRYSMIAN
Draka
General Cable

Cariacica, December 16, 2024.

Marianna Broseguini
Product Development and Methodology Coordinator
Prysmian

To Whom It May Concern,

It is my pleasure to recommend Matheus Ribeiro Vidal for any position requiring knowledge and experience in computational simulation and numerical analysis. I have had the privilege of supervising Matheus during his tenure as an R&D Engineer at Prysmian Group, where he consistently demonstrated strong technical capacity related to the simulations and computational activities, always searching for possibilities of optimization, developing tools to increase the performance of team. He also learned very fast about the product, which was very important, once he become capable to understand the real scenario of application, applying this knowledge on simulations.

In his role, Matheus worked on challenging projects involving simulations for submarine umbilical cable systems, contributing significantly to numerical analyses and design procedures. He learned and successfully utilized tools like OrcaFlex, ANSYS, and Altair Flux 2D to perform simulations that were critical to product development. Moreover, his programming expertise in Python allowed him to optimize internal processes and develop innovative tools that enhanced our methodologies.

One of Matheus's key strengths is the ability on tools developments and the quick learning correlating the simulation area with practical area. He not only executed simulations but also provided theoretical insights and ensured alignment with international standards, which proved invaluable during testing and product development phases.His capacity to bridge theoretical knowledge with practical application is rare and sets him apart as a talented professional in the field of computational engineering.

Matheus's passion for numerical simulation is evident in his work. He consistently showed initiative by seeking innovative solutions and staying updated with the latest advancements in simulation sciences. His strong communication skills and collaborative nature made him an integral part of our team, ensuring seamless coordination across projects.
I am confident that Matheus's skills, dedication, and professional demeanor will make him an asset to any organization. Should you require further information, please do not hesitate to contact me at marianna.broseguini@prysmian.com or +55 027 98182-2436.

Sincerely,
Marianna Broseguini
Product Development Coordinator
Prysmian Group

Broseguini Marianna [brosema001]

Digitally signed by Broseguini Marianna [brosema001]
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To Whom It May Concern,

I am writing to offer my recommendation for Matheus Vidal, who has demonstrated exceptional skills and potential during their work at Prysmian. As a Senior Specialist, at the time, in the Simulations teams at Prysmian, I had the privilege of supervising Matheus on several projects related to numerical simulations and computational modeling. Based on my experience working closely with him, I am confident that he will be an excellent addition to any engineering team.

Throughout his job, Matheus demonstrated his technical proficiency, particularly in the areas of numerical analysis, computational modeling, and simulation software. He quickly adapted to the complex engineering problems we were tackling, applying a range of simulation techniques to generate accurate models and optimize our processes. For example, he developed optimization tools thru python codes, which leaded to increase of performance in process and also improved the accuracy of those process. This contribution significantly enhanced our project outcomes and was a testament to his analytical capabilities.

Matheus also demonstrated strong programming and technical skills, particularly in tools like Ansys and other particular finite element method software such as Orcaflex. His ability to quickly grasp new software and adapt to evolving project requirements was particularly noteworthy.

In addition to his technical skills, Matheus is a strong communicator and a highly effective collaborator. He actively engaged with our multidisciplinary team, offering valuable insights during team meetings and collaborating with colleagues from different backgrounds.

Based on his performance and the qualities he displayed, I am confident that Matheus has the skills, determination, and potential to make a significant impact in the field of numerical simulation engineering. I highly recommend him for any role in this domain and believe that he will be an valuable to your team.

Please feel free to contact me if you have any questions or need further information. I would be happy to discuss Matheus's qualifications in more detail.

Sincerely,
Filipe Cestari Coutinho, M.Sc
Lay Engineer / Offshore Engineer
TechnipFMC
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+55 27 99919-2105
December 13th, 2024